



Jeen (Talk) - AI Voice Agent Home Assignment

Estimated completion time: 8-10 hours

This home assignment is designed specifically for the Junior AI Solution Engineer role for the Jeen Talk team.

The assignment is focused on evaluating practical thinking and hands-on implementation abilities related to Voice AI systems, product thinking, AI integrations, and technical problem-solving.

The assignment is divided into 2 main parts:

- Voice AI Agent & Product Thinking
- Python & AI Integration Task

There is no single "correct" solution - we are mainly looking to understand your thinking process, technical approach, creativity, and decision-making.

You are welcome to use AI tools and assistants throughout the assignment process, including for research, brainstorming, prompt writing, coding, debugging, and solution development.

The goal is not to avoid AI tools, but to understand how you work with them and how you approach problem-solving.

If you have any questions during the assignment, feel free to contact us directly at any time through the recruiter or hiring team.

Good luck!

Part 1 - Business Thinking & Voice AI Agent

1.1 Choose a known B2C company from the Israeli market.

Examples: insurance, banking, healthcare, retail, food delivery, travel, etc.

1.2 Describe 2 possible Voice AI use cases that can improve the customer experience. One Inbound and One Outbound.

1.3 Choose one of the use cases and explain:

- The business value
- The expected ROI
- Why Voice AI is a good solution for this use case

1.3.1 Perform a comparison between different voice agent model architectures based on the models and technologies you selected.

You may compare up to 6 models in total (you may also choose fewer if preferred).

The comparison should be performed between equivalent model categories only.

For example:

- STT vs STT
- LLM vs LLM
- Any additional technologies you believe are relevant

The comparison should focus on the parameters you believe are most important for building a high-quality Voice AI agent.

Explain the advantages and disadvantages of each architecture and which approach you would choose for the selected use case and why.

1.4 Build the selected use case as a working outbound Voice AI agent using any Voice AI platform of your choice, based on the models you selected.

Supported platforms (examples): Vapi, Retell AI, Bland AI, or any similar platform. You may use the free credits provided by the platform.

Please note that most of the supported platforms provide approximately \$10 of free credits for testing purposes.

Use the credits wisely - you are not expected to spend beyond the free credits for this assignment.

Voice Agent Requirements

- 1.4.1** The Voice AI agent must speak Hebrew only.
- 1.4.2** The agent should simulate or perform an outbound phone call flow.
- 1.4.3** There is no requirement to connect a real phone number.
You may demonstrate the outbound call using the built-in test interface / simulator provided by the platform.
- 1.4.4** If you prefer, you may connect a real phone number, but this is optional and not required for the assignment.
- 1.4.5** Use at least 3 dynamic variables during the conversation flow.
Examples: customer name, customer history, requested action, appointment date, insurance type, etc.
- 1.4.6** The variables should be pulled dynamically from Google Sheets, Airtable, webhook, Mirror Table, or mock JSON data.
- 1.4.7** At the end of the conversation, the agent should generate a short conversation summary and write the summary back into the same Google Sheet / Airtable / data source used during the call.
- 1.4.8** The submission must include a summarized table with at least 3 example conversations / rows.
At least one of the rows must include a successful outbound conversation example.
- 1.4.9** Submit one recording or simulator playback of the outbound conversation in Hebrew.

Presentation Requirements

- 1.5.1** Prepare 4-6 presentation slides in Hebrew or English.
- 1.5.2** The presentation should include:
 - The 2 use cases
 - The selected use case
 - The models comparison
 - The selected models and the reasons for choosing them
 - The agent flow and architecture
 - The expected business value

Part 2 - Python Task

- 2.1 Build a simple Python script that receives a text file or PDF and creates embeddings for the content.
- 2.2 Extract text from a TXT or PDF file.
- 2.3 Split the text into chunks.
- 2.4 Create embeddings using OpenAI or Gemini API.
- 2.5 Store the chunks in memory or in a simple local database/file.
- 2.6 Create a simple search function that receives a question and returns the most relevant chunk.
- 2.7 Use a .env file for API keys.
- 2.8 Upload the project to GitHub with a README file.

Submission Requirements

- 3.1 Presentation (PDF or PPTX).
- 3.2 GitHub repository.
- 3.3 One recording or simulator playback of the outbound conversation in Hebrew - MP3/MP4.
- 3.4 Prompt file of the agent.
- 3.5 JSON export of the agent (if possible).
- 3.6 Google Sheets, Airtable, webhook, Mirror Table, or mock JSON data used in the flow.
- 3.7 Screenshots of:
 - The agent configuration
 - The flow
 - The summarized table/results
- 3.8 README file with setup instructions.

Evaluation Criteria

- Product thinking
- Prompt quality
- Voice UX
- Code quality
- AI understanding
- Creativity

Good luck!